

Agent Performance Scorecard

Activity is not an outcome. Outcomes are not a control.

Product	AP Agent OS
Tier	Professional
Version	1.0
Date	September 2026
Use	Implementation toolkit

Responsibility is earned.

Four families

Do not mix these on one vanity tile.

Family	Question	Examples
Activity	Did the agent work?	Invoices handled, exceptions touched, drafts prepared
Operational	Did the process improve?	STP, cycle time, exception resolution, ageing, on-time follow-up
Financial	Did economics move?	Cost per invoice, hours released, validated savings, inference cost
Risk / control	Did we stay safe?	Control breaches, audit exceptions, false negatives on high-risk codes, override rate

Definitions and formulas

Let N be items in the measurement window. Use the same window for numerator and denominator.

Activity

Metric	Formula	Notes
Invoices handled	Count of invoices with an agent output	Not a success metric
Exceptions handled	Count of exceptions with a classification or draft	Pair with recode rate
Human intervention rate	Interventions / handled	Intervention = edit or reject

Quality (mostly operational + control)

Metric	Formula	Notes
Classification accuracy	Correct codes / sampled codes	Against gold or dual-human
Extraction accuracy	Fields correct / fields sampled	Field-weighted; amounts weigh more
Matching explanation accuracy	Accepted explanations / sampled	Not ERP match rate
False-positive rate	$FP / (FP + TN)$ or, for queues, rejected candidates / candidates	State which
False-negative rate	$FN / (FN + TP)$	Critical for duplicates, bank, tax
Rework rate	Items returned to agent queue / handled	After human reject

Operational outcomes

Metric	Formula	Notes
STP rate	Invoices posting without human touch / invoices	Define "touch"
Exception resolution rate	Closed in SLA / exceptions	Need taxonomy

Metric	Formula	Notes
Repeat exception rate	Exceptions with same supplier+code in 90 days / exceptions	Root-cause signal
Average resolution time	Sum of hours from create to close / n	Clock stops on hold if policy says so
Time to invoice posting	Receipt → post	Align to Ardent "cycle" only if you measure the same way
On-time supplier follow-up	Sent within SLA / required	Draft ≠ sent
On-time internal follow-up	Same	
Ageing reduction	Δ of \$ or count >X days vs baseline	Always vs baseline
Payment-on-time rate	Paid on or before terms / paid	Agents do not pay
Missing-receipt reduction	Δ open GRNI lines vs baseline	
PO compliance improvement	Δ PO-backed % or Δ EX-PO-*	

Financial outcomes

Metric	Formula	Notes
Cost per invoice	AP operating cost / invoices	Be honest about allocations
Cost per exception resolved	Exception labour + tool / resolved	
AI inference cost	Invoiced model + tool cost	
Cost per correct outcome	(Labour_agent_ops + inference) / correct outcomes	Prefer this to "cost per run"
Estimated human hours released	Baseline minutes – current minutes, × volume	Label estimated until validated
Validated financial savings	Savings with a finance-approved method	Discounts captured, late fees avoided, FTE not backfilled — pick a method and stick to it

Risk / control outcomes

Metric	Formula	Notes
Control breaches	Count of failed key controls	Including SoD and unapproved send
Escalation rate	Escalated / handled	Spike can be healthy
Audit exceptions	IA findings related to agents	
Duplicate invoices detected	Confirmed duplicates / candidates	Precision matter
Duplicate payments prevented	Confirmed would-have-paid, after human action	Rare; do not forecast

Benchmarks (orientation only)

Ardent Partners State of ePayables 2025 (n=204, June 2025): average cost \$9.84; cycle 8.2 days; exceptions 18.4%; STP 35.4%; Best-in-Class cost \$2.65 and cycle 2.9 days. These are not your targets until you measure yourself the same way.

Scorecard cadence

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- Daily: queue, stalls, high-risk codes, incidents
 - Weekly: accuracy sample, hours, inference cost, top codes
 - Monthly: financial + control pack to sponsor
 - Autonomy review: only on evidence pack

Vanity list (do not report to a CFO as success)

- "Prompts written"
- "AI invoices processed" without quality
- Vendor-stated capture accuracy
- Tokens consumed
- Number of agents live

This material is an implementation toolkit. It is not legal, tax, accounting, audit or investment advice. Agents do not authorise payments. Humans remain accountable. Statistics are cited in the Research Ledger.